



Bioactivity-directed evaluation of fruit of *Kigelia africana* (Lam.) benth. Used in treatment of malaria in Iwo, Nigeria

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ARTICLE INFO

Keywords:

Ethno-medicinal survey
Kigelia africana
Antiplasmodial
Isolated compounds

ABSTRACT

Ethnopharmacological relevance: The ancient people of Iwo communities consisting of Ile-Ogbo, Olupona, Iwo and Ogbagba continue to engage in the traditional use of medicinal plants for the treatment and management of common diseases especially malaria.

Aims of this study: This study conducted an ethnomedicinal survey of plants used to treat malaria and feverish conditions by the people of Iwo, Nigeria. It also evaluated the antiplasmodial activity of the morphological parts of *Kigelia africana* (Lam.) Benth., and isolated, as well as characterised pure compounds from the semi-purified fractions of the fruit extract.

Materials and methods: The ethnomedicinal survey was conducted using semi-structured questionnaires administered to only herb sellers in Iwo, Ile-Ogbo, Olupona, and Ogbagba areas of Osun State. Extracts of *K. africana* morphological parts; leaf, root, stem bark, and fruit were obtained by cold maceration in methanol, followed by assessment of acute toxicity (LD₅₀) and antiplasmodial activity in *Plasmodium berghei* infected rats using the 4-day suppressive test model. The most active fruit extract was further subjected to activity-guided fractionation and purification using *n*-hexane, dichloromethane, ethyl acetate (EtOAc), *n*-butanol (*n*-BuOH), and methanol (MeOH) in gradients to obtain the semi-purified fractions and two pure isolated compounds using various chromatographic and spectroscopic techniques.

Results and discussion: From the survey, thirty-one plant species were identified for treating malaria in Iwo area. *Azadirachta indica* leaf was the most frequently used (78.3% of the respondents) while *Manihot esculenta* leaf (3.33%) was the least. The identified plants are distributed among 24 families, with Anacardiaceae and Asteraceae (11.67% each) been the most occurring families. *Kigelia africana* (Bignoniaceae) ranked the 6th position with 60% frequency of occurrence. The LD₅₀ values obtained for the extracts were greater than 5000 mg/kg (p.o.). The chemo-suppression activity of the extracts at 125 mg/kg was in the order of stem bark (26.59%), leaf (41.75%), root (43.95%), and fruit (54.54%). The semi-purified methanol fraction of the fruit showed the most antiplasmodial activity with a percent chemo-suppression of 69.94 and yielded 4-(2,3-dihydroxypropoxy)-3,5-dihydroxy-5-methylfuran-2-one and sucrose.

Conclusion: The use of herbs and medicinal plants either singly or in combination for the treatment of malaria among the people of Iwo community in Nigeria is still well practised. Lack of formal education among most of the respondents and use of same local name for different plants species or plant parts; which often lead to wrong plant collection were among the constraints encountered. *Kigelia africana* has antiplasmodial activity in the order of fruit > root > leaf > stem bark.

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<https://doi.org/10.1016/j.jep.2020.113680>

Received 15 October 2020; Received in revised form 23 November 2020; Accepted 5 December 2020

Available online 8 December 2020

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1. Introduction

Malaria has been ranked the deadliest vector-borne disease and it continues to ravage many parts of the world. Annually, it is the cause of death of over 1.2 million people (World Health Organisation, 2019a). Although malaria is a preventable and curable disease, it is one of the diseases that cause the frequent outpatient visit to hospitals, high maternal mortality. There has been a noticeable increase in the estimated incidence and deaths due to the disease globally from 2015 to 2017. An estimate of 212 million cases was reported in the year 2015 which has increased with almost 7 million in the year 2017. The reported death within these years has also increased with approximately 6000. The percentage of the reported cases and death due to malaria in Africa has remained within the range of 90–93% (World Malaria Report, 2017; WHO, 2019). Nigeria has been reported to have the highest number of malaria cases in Africa (Dawaki et al., 2016). One of the greatest problems facing the eradication of malaria in Nigeria and other African countries is resistance (World Health Organisation, 2015). Other factors that resuscitated the use of medicinal plants include the high cost of conventional drugs, counterfeiting, and some known side effects which are believed to be absent while using herbal preparations.

The practice of the use of plants to treat different diseases by man is as old as man himself. Medicinal plants are selected based on instinct, the doctrine of signature, observations, and so on. As a result, this study conducted, an ethnomedicinal survey to identify and document such medicinal plants used by the people in the Iwo area of Osun State, Nigeria in the treatment of malaria and to validate the antiparasmodial activity of *K. africana* and to identify constituents of the active plant part.

Kigelia africana (Lam.) Benth., commonly called the sausage (name derived from the shape of the fruit) or cucumber is known in Yoruba ethnomedicine as Pandoro. It is a monospecific genus of the Bignoniaceae family which grows along watercourses and in riverine areas (Grace et al., 2002). It is found distributed in South, Central, and West Africa. All parts of the plant are used in ethnomedicine to treat nasopharyngeal and skin infections, haemorrhoids, malaria (Saini et al., 2009), wounds, rheumatism, etc. (Burkill, 1985; Owolabi et al., 2008; Oyelami et al., 2012). Some of the pharmacological activities attributed to the plant include central nervous stimulant (Owolabi et al., 2008), diuretic activity (Atawodi and Olowoniyi, 2015), haematological parameters (Oyebanji et al., 2015), fertility enhancement (Dada et al., 2010) amongst others. Phytoconstituents of the class of iridoids, flavonoids, naphthoquinones, terpenes and terpenoids have been detected and isolated from different parts of *K. africana* (Bello et al., 2016).

2. Materials and methods

2.1. Administration of questionnaire

An open-ended questionnaire was administered to herb sellers (both sexes) in various markets in four Iwo communities of Ile-Ogbo, Olu-ponna, Iwo and Ogbagba, South-Western Nigeria (Fig. 1). The respondents were asked questions on causes, frequency and different treatments of malaria disease, specific herbs used in treating their patients, and the methods of preparation. The questionnaire was translated from the English Language to the local language (Yoruba Language) of the people for easy communication. Due to the inclusive criterion of the study and the small population of this specific group (herb sellers), only 15 respondents were selected in each locality. The study protocol was approved by the Board of Postgraduate College, OAU, Ile Ife with the registration number PHP15/16/H/2622.

2.2. General laboratory procedure

All organic solvents used were distilled before use and they include methanol, dichloromethane, ethyl acetate, n-hexane, and n-butanol. Vanillin/sulphuric acid (1% w/v) was the spray reagent. The stock of the Giemsa stain was diluted to 5% with Phosphate Buffer Saline. The stationary phases include Silica gel for VLC (100–200 mesh size), Silica gel for Column Chromatography (high-purity grade, Merck Grade 9385), Pore Size 600A, 230–400 mesh particle size, Silica gel for TLC (Silica Gel-G for Thin Layer Chromatography, (with 13% Sulphate as a binder, SQ Laboratory Reagent) for prep-TLC.

Samples (fractions and isolates) were dissolved in appropriate solvents and analysed by TLC using precoated TLC plates followed by development in an appropriate solvent system. The developed plates were allowed to air dry. All TLC plates were first visualised under the UV lamp at 254 and 366 nm before spraying with vanillin/H₂SO₄ reagent followed by heating to reveal spots. Subsequently, fractions were bulked based on the similarity of the TLC profile.

2.3. Collection and extraction of plant materials

For the preliminary antiparasmodial screening, different morphological parts; leaf (203 g), fruit (225 g), stem bark (224 g), and root (120 g) of *K. africana* were collected at Station Road, Iwo, Nigeria. The plant was identified and authenticated by Mr. Ifeoluwa Ogunlowo, taxonomist, Department of Pharmacology, OAU while a voucher specimen IFE-17726 was deposited at the University Herbarium, Department of Botany, OAU, Ile-Ife. The fresh plant parts were cut into smaller pieces and pulverised separately using a manual laboratory grinder. The plant materials were separately extracted by maceration in methanol for 72 h.

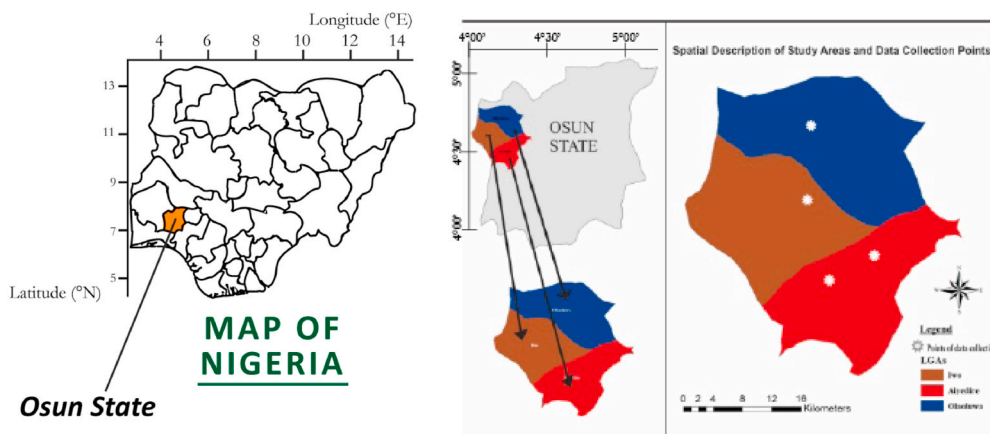


Fig. 1. Spatial description of study areas and data collection points.

The menstrum was continuously agitated on a mechanical shaker during the period of extraction. The extract was filtered and concentrated to dryness *in vacuo* using a rotary evaporator at 40 °C.

2.4. Biological tests

2.4.1. Determination of median lethal dose

The median lethal dose (LD₅₀) of the individual extract was determined according to the method of Lorke (1983). Mice (n = 18) with an average weight of 20 g were used and divided into six groups I - VI (n = 3/group) for each extract. Phase I test uses doses of 10, 100, 1000 mg/kg of each extract, and administered orally to groups I, II, III respectively. Animals in groups I - III were then observed for 24 h for any sign of toxicity such as jerking, writhing, restlessness, mortality, and so on. In phase II, doses of 1600, 2900, and 5000 mg/kg of each extract were administered orally to animals in groups IV, V, VI respectively following the outcome of the phase I experiment. The animals were equally observed for another 24 h for signs of toxicity. The LD₅₀ value was determined using the formula (Enegide et al., 2013):

$$LD_{50} = \sqrt{(D_0 \times D_{100})}$$

D₀ = Highest dose that gave no mortality; D₁₀₀ = Lowest dose that produced mortality

2.4.2. Plasmodium berghei parasite and experimental animals

The test parasite was *Plasmodium berghei*, strain NK 65, obtained from the Institute of Medical Research and Training (IMRAT), University College Hospital, Ibadan, Nigeria. The parasite was maintained by serial passaging and inoculated intraperitoneally.

Experimental animals (both sexes) weighing between 20 and 30 g were obtained from the Animal House, Department of Pharmacology, Faculty of Pharmacy, Obafemi Awolowo University (OAU), Ile-Ife. The animals were kept in the laboratory under standard conditions, fed with standard animal feeds prior to, and throughout the period of experimentation. Protocols for responsible experimentation were strictly adhered to in compliance with the international principles for laboratory Animal Use and Care (Louhimies 2003; Heinrich et al., 2020) and as approved by the Board of Postgraduate College, OAU for animal experiments.

2.4.3. Preparation of inoculum and antiplasmodial assay procedure

Parasitized blood sample was obtained from a donor mouse through cardiac puncture and mixed with an anticoagulant (trisodium citrate, 0.1 ml). Dilution of the blood was based on the number of experimental animals to be inoculated.

Extracts were screened for antiplasmodial activity using the method described by Knight and Peters (1980) and Odeghe et al. (2012). Mice were divided into five groups I - V (n = 5) and were infected by inoculating with 0.2 ml of 1 × 10⁷ P. berghei NK 65 parasitized erythrocytes. Animals in groups I, II, and III were administered oral doses of 125, 250, and 500 mg/kg of each extract respectively, while the semi-purified fractions were accessed as a single dose (250 mg/kg). The positive control group IV was treated with 10 mg/kg standard chloroquine while the negative control, group V, received 0.2% DMSO (for extracts test) and distilled water (for fractions test). Treatments were done for four consecutive days. On the fifth day, blood collected from the tail of each mouse was made into thin smear and stained with Giemsa stain for microscopic determination of the level of parasitemia in each animal. The percentage parasitaemia and chemosuppression of the extracts were then determined. Following the preliminary antimalarial screening of all the morphological parts of *K. africana*, the fruit with the best antiplasmodial activity was re-collected in bulk further bioassay evaluation and phytochemical investigation.

2.5. Phytochemical study of the fruit of K. africana

The fruit (3.6 kg) was re-collected from the same location and processed as earlier described. The pulverised fruit was macerated in MeOH (4.3 L) for 72 h with constant agitation on an electric laboratory shaker. The menstrum was thereafter filtered and concentrated to complete dryness *in vacuo* at 400 °C. The extract was successively fractionated with *n*-hexane, dichloromethane, ethyl acetate (EtOAc), *n*-butanol (*n*-BuOH), and methanol (MeOH) using Vacuum Liquid Chromatography. The LD₅₀ and antiplasmodial effects of the EtOAc, *n*-BuOH, and MeOH fractions were assessed as described earlier. Based on the result of the antiplasmodial assay, the MeOH fraction with the best activity was subjected to further purification by VLC using EtOAc: MeOH solvent mixtures in gradient to yield three main sub-fractions A (7.12 g), B (9.44 g), and C (330 mg) based on their TLC characteristics in solvent system *n*-BuOH: Acetic acid: H₂O (15: 0.1: 1). Sub-fraction A (7.12 g) eluting with EtOAc: MeOH (30: 70–100% MeOH) was separated by column chromatography. It was pre-adsorbed on silica gel (100–200 mesh size) and eluted using EtOAc: MeOH mixture in gradient to yield sub-fractions A1 (561 mg) and a pure compound A2 (335 mg). Sub-fraction A1 eluting with EtOAc: MeOH (40: 60–30: 70) was further purified by PTLC (*n*-BuOH: Acetic acid: H₂O 15: 0.1: 1) to afford partially purified fractions A13 (23 mg) and A14 (16 mg). Fraction B (9.44 g) which eluted with EtOAc: MeOH (70: 30–40: 60) was separated by VLC to give sub-fractions B1 (310 mg) and B2 (5.4 g). Purification of B1 was achieved using PTLC (*n*-BuOH: Acetic acid: H₂O 15: 0.1: 1) to yield partially purified fractions B11 (3 mg) and B13 (19 mg) and a pure compound B12 (22 mg).

2.6. Statistical and spectroscopic analyses

The graphs, calculations, and statistical analyses including the mean, standard deviation, and standard error of mean values were performed using Microsoft office excel.

The spectroscopic analysis carried out on the isolates includes ¹H NMR, ¹³C NMR, DEPT, COSY, and HSQC on a Bruker 500 MHz Spectrometer. Electrospray Ionization Mass Spectrometry on a Thermo Quest Finnigan LC QDECA Spectrometer.

3. Result

3.1. Ethnomedicinal survey

A total of 60 respondents were interviewed, 15 from each town of Iwo, Ile-Ogbo, Oluponna, and Ogbagba, Osun State, Nigeria. The respondents consist of 90% female and 10% male, aged <30 representing 13.3% and >30 years representing 86.7%. Most of the respondents had no formal education (90%), 8.3% had secondary school education while 1.7% had tertiary education. All the respondents were herb sellers.

A complaint of high body temperature and headache (100%) were the major symptoms of malaria indicated by the respondents, followed by body pains (95%). Vomiting (36.7%) and sleeplessness (21.7%) were also identified as symptoms of malaria. Body weakness was identified by only 10% of the respondent as a possible symptom of malaria.

A higher percentage of the respondents (90%) attributed that mosquito vector contributes largely to the transmission of malaria fever. Other causes identified by the respondents include lack of proper environmental sanitation (23.3%) and lack of potable water (38.3%). Sun, a gift from nature, was also attributed as a cause of malaria by 6.7% of the respondents. Constant exposure to sun radiation was said to lead to high temperatures and then malaria. The effect of too much stress was also mentioned by 3.3% of the respondents.

The occurrence of malaria fever was opined to be all year round by 38.3% of the respondents, while 43.3% claimed that malaria is commonest during the wet season (March–October). However, 18.3% of the respondents said that malaria infection is predominant during the dry

season (November–February).

Preventive measures prescribed by the people include avoidance of stagnant pool of water in the environment (78.3% of the respondents), cutting the lawn, grasses, and bushes in the surroundings (86.7%), proper sanitation and cleanliness of the environment, and proper disposal of unused cans, bottles and containers (21.7%). Netting of houses was attributed by 96.7% of the respondents as a great way and the most effective way of combating malaria. All of the respondents are aware of Insecticide Treated Net (ITN) but 71.7% have used ITN at one time or the other.

A total of 31 different plant species belonging to 24 families of plants were identified from the survey (Table 1). Different morphological parts such as leaf, bark, root, seed, and fruit are used by the people either singly or in combination to treat malaria. The most occurring family of plants from the survey are Anacardiaceae and Asteraceae, others include Malvaceae, Bignoniaceae, Rutaceae, Leguminosae, Apocynaceae, Poaceae, Meliaceae, Annonaceae, Rubiaceae, Fabaceae, Lamiaceae, Cucurbitaceae, Lythraceae, Myrtaceae, Ochnaceae, Amaranthaceae, Piperaceae, Mimosaceae, Caricaceae, and Euphorbiaceae. The use of some of the plants for the treatment of malaria has been validated scientifically. For instance, the antimalarial activity of *Azadirachta indica* leaf was reported by MacKinnon et al., 1997, *Eupatorium odoratum* leaf (Runghana, 2003), *Acanthospermum hispidum* leaf (Granfon et al., 2012), *Morinda lucida* leaf (Richard et al., 2006), *Enantia chlorantha* bark (Vennerstorm and Klayman, 1988) amongst others.

3.2. Median lethal dose and anti-plasmodial effects of *K. africana* parts

The median lethal dose (LD₅₀) of the different morphological parts; leaf, stem bark, root, and fruit extracts of *K. africana* were greater than 5000 mg/kg. No sign of toxicity, behavioural changes, or death was observed in all animal post-administration of the extracts.

All extracts were evaluated for 4-day antiparasmodial testing in mice and showed dose-dependent chemosuppression activity at the three test doses of 125, 250, and 500 mg/kg. The positive control (chloroquine) had a % chemosuppression of 93.99. The stem bark extract gave the highest chemosuppression activity of 84.73%, followed by the root (78.06%) and then the leaf (72.94%) while the fruit had the least activity of 70.33% at the maximum dose of 500 mg/kg. However, it was observed that the fruit extract had the best activity of 54.54%, followed by the root (43.95%), leaf (41.75%), and the stem bark had the least activity of 26.59% at the lowest dose of 125 mg/kg (Fig. 2).

3.3. Median lethal dose and anti-plasmodial effects of solvent fractions of *K. africana* fruit extract

The EtOAc and *n*-BuOH solvent fractions of the fruit extract had LD₅₀ greater than 5000 mg/kg while the MeOH fraction had LD₅₀ of 3807.9 mg/kg. The *n*-hexane and DCM fractions were not tested due to the small yield of the fractions. Percentage chemosuppression of EtOAc, *n*-BuOH, and MeOH fractions are 42.61, 56.05, and 69.94 respectively while chloroquine had a chemosuppression of 86.17% (Fig. 3).

3.4. Isolation and purification of MeOH fraction

Compound B12 (22 mg) was obtained as a brown coloured powder material which was highly polar (soluble in MeOH and water) with R_f value of 0.48 (*n*-BuOH: Acetic acid: water/15: 0.1:1), developed yellow colouration to Vanillin/H₂SO₄ (V/S) spray after heating. Proton and carbon-13 NMR data are presented in Tables 2 and 3 respectively. The ¹³C spectrum showed the presence of eight highly deshielded signals, while the DEPT spectrum showed six signals indicating the presence of two quaternary carbon atoms. The presence of two methylene (CH₂) signals was observed at δ 61.71 and δ 64.87. Three methine (CH) signals found at δ 79.00, δ 83.66 and δ 77.37 were assigned to C-21, C-4, and C-3 respectively. Additionally, the presence of a methine (CH) (δ 83.66)

Table 1

Medicinal plants used by the people of iwo area for the treatment of malaria.

S/ N	LOCAL NAME	SCIENTIFIC NAME	FAMILY	PART USED	% USED
1.	Dongoyaro	<i>Azadirachta indica</i> (A. Juss)	Meliaceae	Leaf	78.3
2.	Akintola-ta- ku	<i>Eupatorium</i> <i>odoratum</i> (L.)	Asteraceae	Leaf	75
3.	Dagunro	<i>Acanthospermum</i> <i>hispidum</i> (DC.)	Asteraceae	Leaf	70
4.	Awopa	<i>Enantia chlorantha</i> (Oliver)	Annonaceae	Bark	68.3
5.	Ooruwo	<i>Morinda lucida</i> (Benth.)	Rubiaceae	Leaf and Root	68.3
6.	Pandoro	<i>Kigelia africana</i> (Benth.)	Bignoniaceae	Root	60
7.	Imi-esu	<i>Ageratum</i> <i>conyzoides</i> (L.)	Asteraceae	Leaf	58.3
8.	Ahun	<i>Alstonia boonei</i> (De Wild.)	Apocynaceae	Bark	56.7
9.	Kaju	<i>Anacardium</i> <i>occidentale</i> (L.)	Anacardiaceae	Bark	51.7
10.	Kaju	<i>A. occidentale</i> (L.)	Anacardiaceae	Leaf	40
11.	Banni	<i>Prosopis cineraria</i> (L.)	Fabaceae	Bark	38.3
12.	Efinrin/ Efinrin aja	<i>Ocimum gratissimum</i> (L.)	Lamiaceae	Leaf	36.7
13.	Ogbo	<i>Parquetina</i> <i>nigrescens</i> (Afzel.)	Apocynaceae	Leaf	36.7
14.	Pandoro	<i>Kigelia africana</i> (Benth.)	Bignoniaceae	Bark	36.7
15.	Ejirin	<i>Momordica</i> sp. (L.)	Cucurbitaceae	Leaf	35
16.	Mangoro	<i>Mangifera indica</i> (L.)	Anacardiaceae	Bark	31.7
17.	Omisinmisin	<i>Abrus precatorius</i> (L.)	Leguminosae	Leaf	31.7
18.	Kooko oba	<i>Cymbopogon</i> <i>citratrus</i> (D. C.) Stapf	Poaceae	Leaf	28.3
19.	Laali	<i>Lawsonia inermis</i> (L.)	Lythraceae	Leaf	28.3
20.	Owu	<i>Gossypium</i> <i>arborescens</i> (L.)	Malvaceae	Leaf	28.3
21.	Kanafuru	<i>Syzygium</i> <i>aromaticum</i> (L. Merry et. Perry)	Myrtaceae	Seed	26.7
22.	Egbesi	<i>Nauclea latifolia</i> (Smith)	Rubiaceae	Bark and leaf	25
23.	Oganho	<i>Khaya</i> <i>grandifoliola</i> (Welw.)	Meliaceae	Leaf	25
24.	Ponhan	<i>Lophira alata</i> (Banks ex Gaertn)	Ochnaceae	Leaf	25
25.	Tete elegun	<i>Amaranthus</i> <i>spinosus</i> (L.)	Amaranthaceae	Leaf	23.3
26.	Mangoro	<i>Mangifera indica</i> (L.)	Anacardiaceae	Leaf	21.7
27.	Abafe	<i>Piliostigma</i> <i>reticulatum</i> (DC.)	Leguminosae	Leaf	20
28.	Eeru	<i>Xylopia aethiopica</i> (Dun A. Rich)	Annonaceae	Seed	20
29.	Oronbo wewe	<i>Citrus aurantifolia</i> (Christm.)	Rutaceae	Leaf	20
30.	Iyere	<i>Piper guineense</i> (Schum. & Thonn.)	Piperaceae	Seed	18.3
31.	Laali	<i>Lawsonia inermis</i> (L.)	Lythraceae	Root	15
32.	Sepeleba	<i>Tithonia</i> <i>diversifolia</i> (Hemsl.)	Asteraceae	Leaf	15
33.	Aidan	<i>Tetrapleura</i> <i>tetraptera</i> (Schum. & Thonn.)	Mimosaceae	Leaf	11.7
34.	Ibepe	<i>Carica papaya</i> (L.)	Caricaceae	Unripe fruit	5
35.	Ege	<i>Manihot esculenta</i> (Crantz)	Euphorbiaceae	Leaf	3.3

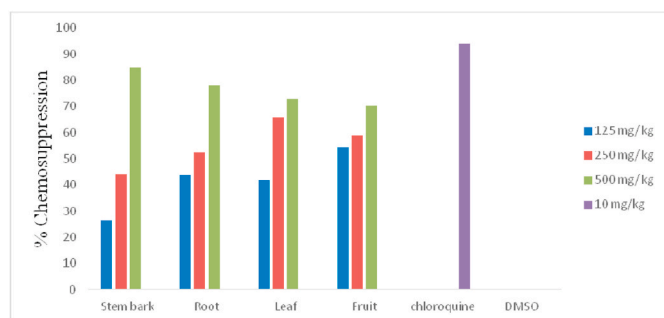


Fig. 2. Antimalarial activity of various morphological parts of *K. africana*.

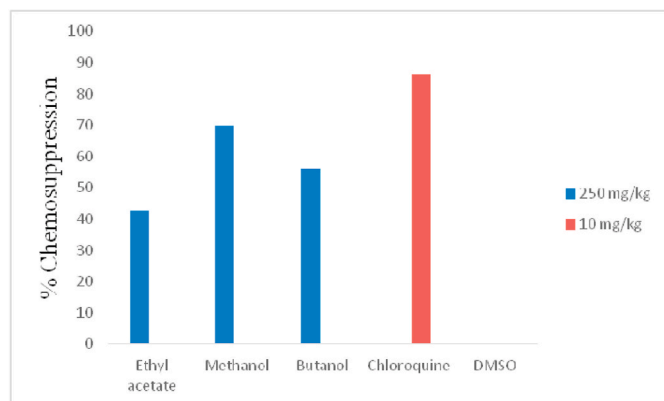


Fig. 3. Antimalarial activity of the fractions of the fruit extract.

Table 2

NMR data (500 MHz; CD₃OD): ¹H NMR and COSY experiments of compound B₁₂.

Position	δ value (ppm)	Multiplicity	COSY
3	4.11	(1H, d; J = 8.0 Hz)	H-4
4	3.94, 3.93	(1H, t, (d); J = 8.0 Hz)	H-1 ¹ , H-3
6	1.89	(3H, s)	-
1 ¹	3.72, 3.75	(2H, d, d; J = 3.5 Hz)	H-4
2 ¹	3.59	(1H, m)	H-1 ¹
3 ¹	3.65, 3.55	(2H, d, d; J = 12.0 Hz)	H-2 ¹

Table 3

NMR data (500 MHz; CD₃OD): ¹³C NMR and DEPT experiments of compound B₁₂.

C position	δ C (ppm)	DEPT
2 (C=O)	180.51	Cq
3	78.99	CH
4	77.37	CH
5	105.34	Cq
6	24.38	CH ₃
1 ¹	64.87	CH ₂
2 ¹	83.66	CH
3 ¹	61.71	CH ₂

and a methylene carbon (δ 61.71) at positions 4 and 11 respectively are very characteristic of C–O–C (epoxide link). The only upfield signal found in the spectrum was a methyl group at δ 24.38 assignable to C-6. No aromatic signal was found in the spectrum. The presence of any type of glycosidation was also not obvious in the spectrum.

The presence of a terminal oxygenated methylene group is evident from the proton signals at δ 3.65 and 3.55 (2H) with a characteristic large germinal coupling (each d, J = 12.0 Hz) (Table 2) was obvious in

the ¹H-NMR spectrum. Protons H-3 (δ 4.11, 1H, d; J = 8.0 Hz) and H-4 (δ 3.93, 3.94, 1H, t (d); J = 8.0 Hz) were readily describable from the spin system involving their correlation with each other by COSY (Table 2). A signal at δ 1.89 integrating for 3 protons was assigned to the methyl group at position-6. Protons appearing at δ 3.72 and 3.75 (d, d; each J = 3.5 Hz) were assigned to the two protons on H-11 and with long-range correlation with H-4 of the epoxide linkage (Table 2).

The mass spectrometry analysis of compound B₁₂ showed M/z 222.99 [M]⁺ corresponding to the molecular formula C₈H₁₄O₇ and a base peak of 149. Other fragment ions found are 204.99 [M-H₂O]⁺; 191.95 [M-CH₂OH]⁺; 161.02 [M-C₂H₅O₂]. Compound B₁₂ was therefore identified as 4-(2,3-dihydroxypropoxy)-3,5-dihydroxy-5-methyl-furan-2-one (Fig. 4).

Compound A₂ has a very light brown colouration, gummy to touch, and also soluble in methanol. It has an R_f value of 0.14 (BuOH: CH₃COOH: H₂O/15:0.1:1), developed yellow colour with v/s spray. Compound A₂ was confirmed to be sucrose after comparing its ¹³C spectrum with the report of Jones et al., 1979. The ¹³CNMR data is presented in Table 4 while the structure is shown in Fig. 5.

4. Discussion

Identifying and documenting the indigenous knowledge of the people across the globe is very crucial to preserve posterity for the coming generations, especially in places where there is no recorded information. An ethnomedicinal survey is a much-known tool for achieving the identification of various medicinal plants used by the people in a selected area. The selected area for this study is predominantly a Yoruba speaking area in South-western Nigeria. The questionnaire administration was in the native Yoruba language of the people since the majority of the respondents had no formal education. The questionnaire was designed using a combined system (open and close-ended questions), which are quite easy to understand and responded to (Mohammed et al., 2017; Kodi et al., 2017; Madara et al., 2018). For instance, direct questions like what time of the year is malaria pre-dominant were responded to as rainy season or dry season or both. About half of the respondents (43.3%) said malaria is mostly predominant during the rainy season. This is a period of the year in Nigeria between months of March to October with low to heavy rainfall. However, some respondents (>20%) claimed that malaria infection occurs mostly during the dry season (November to February), while about 40% of the respondents acclaimed that malaria occurs throughout the year including both rainy and dry seasons. The open-ended system also allowed respondents to give a wide range of responses. For instance, questions such as what medicinal plants are used in treating malaria allowed the respondents to give a list of possible and potential plants instead of limiting the responses. A total of 31 plant species belonging to 22 families were identified by the respondents as potential antimalarial plants. The most occurring families were Asteraceae and Anacardiaceae. A previous study by Olorunnisola et al., 2013 conducted among people in

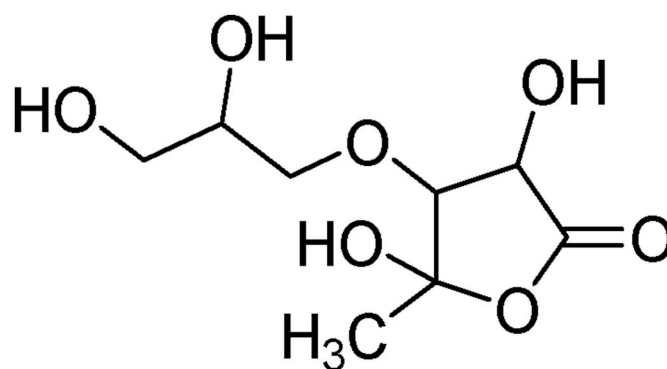
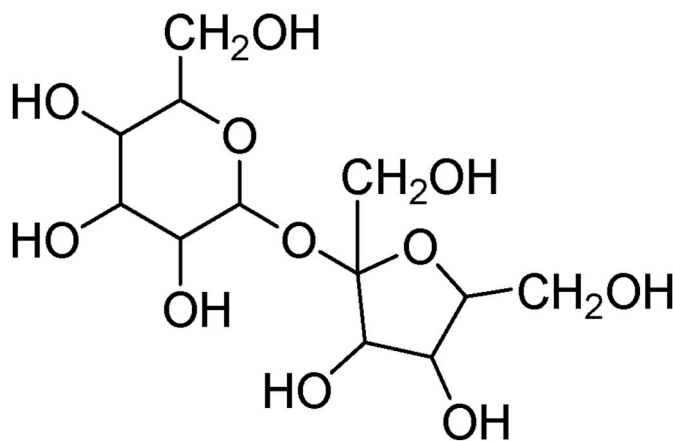


Fig. 4. Structure of compound B₁₂.

Table 4¹³C NMR data of compound A₂ (500 MHz; CD₃OD).

C position	Compound A ₂ (ppm)	Jones et al. (1979)
1	92.2	92.9
2	73.0	71.9
3	74.0	73.4
4	70.0	70.0
5	71.8	73.2
6	60.8	61.2
1 ¹	62.7	62.2
2 ¹	103.9	104.5
3 ¹	77.8	77.3
4 ¹	74.9	74.8
5 ¹	82.4	82.2
6 ¹	62.0	63.2

**Fig. 5.** Structure of compound A₂.

Ogbomoso South, Oyo State, Nigeria also identified and reported these two families as the most represented families in their survey. In another study by Madara et al., 2018, 30 species of plants occurring in 16 families were identified by the Rumaya people of Kauru Local Government Area of Kaduna State with Fabaceae and Combretaceae as the most represented families. The ethnomedicinal study carried out in this study identified *Azadirachta indica* as the most effective plant used in the treatment of malaria by the people in the Iwo area of Osun State. Interestingly, *A. indica* was also reported by Lagnika et al., 2016 to be used by the people of South Benin. It was also mentioned in the report of Awosan et al., 2014 to be one of the most effective medicinal plants used to treat this disease in Akinyele local government area, Ibadan. The leaf (65.7%) was the most used part followed by the stem bark (20%) and then the root and seed (8.6%) while the fruit was the least used (2.9%). The high percentage of use of the leaf could be due to its easy accessibility while the low percentage of use of the seed and fruit could be due to the seasonal occurrence of these plant parts.

The majority of the respondents combined both herbal and orthodox medicines in the treatment of malaria not minding the possibility of drug-herb interaction. As expected, none of them use orthodox medicine alone to treat malaria. A larger percentage (90%) of the respondents understands that mosquito contributes immensely to the transmission of malaria. A few (6.7%) mentioned the sun to be a cause of malaria fever. Lagnika et al., 2016 reported that all the respondents (100%) mentioned mosquito bite to be the cause of malaria, while a much higher percentage (71.4%) of the respondents attributed that working in the sun can also cause malaria. The use of medicinal plants to treat malaria and other infectious diseases is still much practiced by the indigenous people.

Acute toxicity (LD₅₀ determination) has been defined by Enegide et al., 2013 as the unwanted effects that occur either immediately or on a

short time interval after a single or multiple administration of a test substance within 24 h. Acute toxicity testing is necessary for dose determination. All the test extracts had LD₅₀ value > 5000 mg/kg and are therefore relatively safe since no death was recorded and no sign of toxicity was noticed in all the animals. A previous study by Sharma et al., 2010 had reported the LD₅₀ of the aqueous bark extract of *K. africana* as 5000 mg/kg. In another study by Adoum in, the methanol extract of the fruit when administered at 6400 mg/kg to test animals, signs of toxicity for example jerks, writhes and 60% death was noticed and at 12,800 mg/kg, 80% death was recorded hence the LD₅₀ was reported to be 3981.07 mg/kg. The toxicity study by Hayat et al., 2017 recorded no mortality at 50, 500, and 2000 mg/kg. Akanni et al., 2017 reported that the ethanol extracts of the fruit, leaf, and stem bark of *K. africana* when administered orally at 100–6400 mg/kg body weight did not produce significant changes in behaviour of the animals and no mortality was recorded during the period of observation.

The extract of the different morphological parts of *K. africana* showed dose-dependent antiplasmodial activity at all the tested doses of 125, 250, 500 mg/kg. In this study, the stem bark extract of *K. africana* gave the best % chemosuppression of 84.73% at 500 mg/kg followed by the root (78.06%), the leaf (72.94%), and the fruit (70.33%). However, at the lowest dose (125 mg/kg), the fruit extract exhibited the highest percentage chemosuppression of 54.54%, followed by the root with 43.95%, the leaf 41.75%, and the stem bark 26.59%. Therefore, the fruit extract having the best antiplasmodial activity at the lowest dose was further studied in a bioactivity guided fashion to isolate its constituents. Although, there are no wide gaps in the activity of the fruit and root but selecting the fruit is a means of preserving the plant as the bulk collection of the root can lead to the death of the plant. Also, pharmacologically, lower doses are used when active instead of higher doses. The yields of the non-polar fractions (n-hexane and dichloromethane) of the fruit extract were low and were not tested. This indicates the low content of lipophilic compounds in the fruit as compared to the polar fractions. The moderately polar fraction and polar fractions were thus tested for acute toxicity testing. The LD₅₀ of the EtOAc and *n*-BuOH was greater than 5000 mg/kg while that of MeOH was calculated to be 3807.89 mg/kg. The most polar fraction, MeOH, possesses the best antiplasmodial activity of the three fractions tested.

Kigelia africana is a monospecific genus of the family Bignoniaceae which is known to contain iridoids, naphthoquinones, tannins, flavonoids, and terpenes (Gouda et al., 2003). From the preliminary fractionation of the plant's fruit extract, it was observed that the majority of the putative active antiplasmodial compounds reside in the methanol fraction. The TLC examinations of the fractions following repeated chromatographic purification showed the loss of some component and degradation of some other compounds. The chemical study led to the isolation of a compound which is a highly oxygenated compound and it is identified as 4-(2,3-dihydroxypropoxy)-3,5-dihydroxy-5-methylfuran-2-one. This being the first time of isolating this compound, a trivial name of Ifekigelinone was assigned to it. More so, a phenylpropanoid derivative, 6-p-coumaroyl-sucrose, has been isolated from the fruit of the plant (Gouda et al., 2006). The sucrose isolated may be a fraction of this compound and this is the first report on the isolation of sucrose from this plant.

5. Conclusion

The use of herbs and medicinal plants either singly or in combination for the treatment of malaria among the people of Iwo community in Nigeria is still well practised. The study documented 31 plant species with the most occurring plant families identified as Asteraceae and Anacardiaceae. Generally, from the survey, the leaf part is mostly used plant part possibly due to ease of access to this plant part. Lack of formal education among the majority of the respondents, and use of same local name for different plants or parts; which often lead to wrong plant collection were among the constraints encountered. *Kigelia africana* in

particular has antiparasmodial activity in the order of fruit > root > leaf > stem bark.

Authors' contributions

1. Authors CAE and JMA initiated, designed and supervised the work, and coordinated the preparation of the manuscript.
2. Author IZI carried out the biological experiments; mean lethal dose (LD₅₀) and antiparasmodial assay under the guidance of NOO and the phytochemical investigation under the supervision of CAE.
3. Author IIZ carried out the ethnobotanical survey.
4. All authors reviewed and edited the original draft of the manuscript.

Funding sources

The authors did not receive any financial support or specific grant from funding agencies in the public, commercial, or not-for-profit sectors for the conduct of this research and/or authorship of this article.

Acknowledgement

Author CAE wishes to acknowledge Prof. Armando McDonald, Department of Forest, Rangeland and Fire Sciences, University of Idaho, Idaho, USA for NMR and MS analyses.

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